MIL-C-24643/45B
3 October 1986
SUPERSEDING
MIL-C-0024643/45A(SH)
8 November 1985
MIL-C-24643/45
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LS2U (INCLUDING VARIATION LS2UA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS2U (unarmored) and LS2UA (armored)

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, in accordance with ASTM B 286, size 26-7. Maximum diameter 0.020 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification code applied by method 3.
Fourth - Two conductors cabled together to form a pair, with nominal lay of 1-1/2 inches.
Fifth - The specified number of pairs (see table I) cabled together with lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward.
Sixth - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.
Seventh - Braided shield of AWG no. 34 tin coated copper. Minimum coverage of 88 percent.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS2U

Eighth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS2UA

Eighth - Cross-linked polyolefin jacket. (See table I for thickness).
Ninth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/45	Size 2/	Number of pairs	Cable jacket thickness (nominal) (inch)	Overall diameter (inches)			
				Type LS2U		Type LS2UA	
				Minimum	Maximum	Minimum	Maximum
-01UO	10	10	0.050	0.450	0.480	0.500	0.530
-02UO	15	15	.050	.530	.560	.580	.610
-03UO	19	19	.050	.550	.580	.600	.630
-04UO	30	30	.050	.670	.700	.720	.750
-05UO	45	45	.050	.830	.870	.880	.920
-06UO	60	60	.065	.920	.960	.970	1.010

1/ Unarmored, overall shield. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Types LS2U and LS2UA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	45.19
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Insulation resistance - megohms/1000 feet, minimum.....	500
Jacket flaws.....	No failure
Conductor and shield continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Characteristic impedance - at 1 MHz, ohms.....	135 ± 15

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - at minus 30 ± 2°C, maximum mandrel diameter 12 times cable diameter.....	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum.....	25
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing (jacket) type LS2U only - cycles, minimum.....	250
Heat distortion - percent of unaged, maximum.....	30
Cable filler removability.....	No failure
Armor (type LS2UA only) - conformance to material, construction and coverage.....	No failure
Shield - conformance to material, construction and coverage.....	No failure
(B) Surface transfer impedance (type LS2U only)	
Milliohms per meter, maximum.....	700
EMP response time - ms/500 feet.....	60

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-45)

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